

Social Gaze -- Graph(V14.5) x VLM Pipeline

VSGaze multi-person social gaze: per person-pair predict **LAH** (i looks at j, directed), **LAEO** (mutual gaze), **SA** (shared attention). Metrics: **F1_LAH**, **F1_LAEO** (per-target-argmax + thr 0.5), **AP_SA** (per-pair AP). Goal: connect the pretrained **V14.5 gaze graph** (geometric specialist) with a **fine-tuned VLM** (Qwen3-VL-8B) and study what the VLM adds.

1. Pipeline overview

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VSGaze frame + head bboxes | +- V14.5 graph (frozen) --> per-pair gaze beliefs/edges + per-person gaze heatmap | +- export aligned features: results/ssg_cache/v14graph_{val,train,test}.pt | +- pair overlay render --> results/vlm_overlays/<split>/<sid>/<i>_<j>.png | (full frame, query pair boxed i=red / j=blue, GT-labelled pairs only) | v VLM (Qwen3-VL-8B + LoRA) -- two interface variants: (A) graph-TEXT : serialize graph outputs to text, prepend to query (B) graph-TOKEN : inject graph EMBEDDINGS as soft tokens (latent fusion) | v yes/no --> P(yes) --> compute_metrics --> F1_LAH / F1_LAEO / AP_SA
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2. Components & key settings

2.1 Gaze graph (specialist, frozen)

item	value
checkpoint	checkpoints/best_V14.5.ckpt (MTGS-derived, NeurIPS'24 base)
backbone	DINOv2 ViT-B/14 (frozen), conditional DPT gaze-heatmap decoder
temporal_context	2 (5-frame clip)
role	frozen front-end ; provides geometric gaze topology, not retrained

Exported per-sample features (v14graph_{split}.pt[sid], n = frame's #people): v_src (n,256) looker embeddings / v_tgt (n+2,256) target embeddings / edge_pp (n,n,256) directed pair embeddings / edge_null_in/out (n,256) (in-frame-object / out-of-frame) / lah/laeo/sa_logits (n,n) + *_frames (per-frame) / gaze_vecs (n,2) / gaze_point (n,2) (heatmap argmax, norm xy) / overlap (n,n) (heatmap mass in target bbox) / head_bboxes, GT, dataset.

Alignment (critical): features must match the overlays' exact frame/people. Export reproduces the render's DataLoader RNG -- **num_workers: train=10, val/test=8**, _one collate, seed(0) AFTER model load; self-check vs gtmeta_<split>.pt aborts on mismatch (all splits verified 0 bad).

2.2 Data / overlays

item	value
dataset	VSGaze (VAT + ChildPlay + UCO-LAEO + VideoCoAtt)
overlay	full frame, pair-only : query i = red box, j = blue box (no other-person IDs)
pairs	GT-labelled pairs only (ans in {yes,no})
records	val 56,692 / test 866,192 (per (sid,task,i,j))

2.3 Interface A -- graph-TEXT

Serialized per-pair cues, prepended to the query (numeric, person-labelled). Per LAH example:

P1 (red) gaze points left, near horizontal; predicted gaze point (0.20,0.26). Aim at P2 (cos gaze vs P1->P2): +0.99. Gaze-heatmap landing on P2: 0.57. Graph edge score P1->P2: 0.97; P1->nobody/out-of-frame: 0.04. Per-frame P1->P2: 0.41,0.40,0.97,0.97,0.97. Others looking at P2: 2.

Fields: head-box coords, gaze direction(angle), **gaze point (heatmap argmax)**, aim(cos), heatmap overlap, edge score $\text{sig}(\text{lah}[j,i]) + \text{per-frame}$, focal in-degree, null-out/in; LAEO = both directions + mutual; SA = two gaze points distance / ray intersection. (Orientation: $\text{lah_logits}[a,b] = \text{"b looks at a"}$ -- verified by eval.)

2.4 Interface B -- graph-TOKEN (latent fusion)

N_TOK = 10 soft tokens per pair, projected from 256-d graph embeddings into the VLM hidden space and injected at <gtok> placeholders via a forward pre-hook on the text model (after the image merge):

$v_src[i], v_src[j]$ (looker) / $v_tgt[i], v_tgt[j]$ (target) / $edge_pp[i,j], edge_pp[j,i]$ (directed relation) / $edge_null_out[i], edge_null_out[j]$ (out-of-frame) / $edge_null_in[i], edge_null_in[j]$ (in-frame object).

Projector: $\text{Linear}(256 \rightarrow 1024) \rightarrow \text{GELU} \rightarrow \text{Linear}(1024 \rightarrow 4096)$ + per-slot type embedding; trained jointly with LoRA.

2.5 VLM + LoRA (both variants)

item	value
base	Qwen3-VL-8B (checkpoints/qwen3-vl-8b), vision tower frozen
LoRA	rank 16 , alpha 32 (=2 / rank), dropout 0.05 , target = language-model {q, k, v, o, gate, up, down}_proj
trainable	43.6 M (0.50%) [+ projector for graph-token]
precision	bf16; use_cache=False; expandable_segments:True

2.6 Training hyper-parameters

item	value
objective	answer-token cross-entropy on "yes"/"no"
optimizer	AdamW, lr 1e-4 , grad-clip 1.0
batch size	8 (full GPU) / 4 (graph-text on 48 GB MIG, longer prompts), accum 1
epochs	3 , steps_per_epoch 20,000
sampler	WeightedRandomSampler balancing each (task, answer) cell (LAEO=yes ~8% rare)
model select	per-epoch VAL eval (same interface) -> best epoch (graph-text: ep2)
hardware	SLURM gpu:rtx6000 (Blackwell 97 GB) / gpu:mig48gb; venv .venv-mtgs-blackwell
wall-clock	~5 h / 3 epochs (full GPU); per-epoch VAL eval ~50 min

2.7 Evaluation protocol

MTGS/mtgs/performance/compute_metrics.py (locked): **F1_LAH / F1_LAEO = per-target-argmax + thr 0.5; AP_SA = per-pair AP**. Inference saves P(yes) = softmax([yes_logit, no_logit]); P(yes) spliced into MTGS per-sample dicts via the render-pass gtmeta (never re-iterates the dataset -> avoids RNG drift). Test = single unsharded pass (deterministic, 866 k records, ~10 h).

3. Results (F1_LAH / F1_LAEO / AP_SA)

model	F1_LAH (val/test)	F1_LAEO (val/test)	AP_SA (val/test)
graph (V14.5)	0.818 / 0.827	0.774 / 0.742	0.746 / 0.664
zeroshot-vision	0.462 / 0.450	0.510 / 0.406	0.468 / 0.275
zeroshot-graphtext	0.458 / --	0.593 / --	0.570 / --
finetune-vision (no-graph)	0.725 / 0.709	0.700 / 0.617	0.683 / 0.450

finetune-graphtext (ep2)	0.834 / 0.830	0.726 / 0.705	0.806 / 0.620
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graph-text VAL by epoch: ep0 0.826/0.716/0.790 / ep1 0.823/0.679/0.796 / **ep2 0.834/0.726/0.806**.

4. Key findings (so far)

1. **VAL vs graph**: graph-text fine-tune wins LAH (+0.016) and SA (+0.060), **loses LAEO (-0.048)**.
 2. **TEST does NOT transfer**: vs graph on test -> LAH +0.003 (tie), LAEO -0.037, **AP_SA -0.044** (graph-text SA collapses **0.806 -> 0.620**). On held-out test the VLM does **not** beat the specialist (val gains = overfitting).
 3. **Parroting**: on graph-confident-wrong cases the graph-text VLM overrides only **11% (LAH) / 10% (LAEO)**, whereas the same VLM **without** graph-text overrides **37% / 54%** -> conditioning on the graph's answer induces confirmation bias (inherits its confident errors).
 4. **Oracle headroom** (graph<->vision perfect routing, per-target acc): LAH **0.898**, LAEO **0.982** vs realized ~0.828 / 0.948 -> large unrealized complementarity.
 5. **VLM alone < graph** on all three (0.725/0.700/0.683) -- the VLM does not surpass the specialist on the standard relations; its value (if any) is complementary.
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5. In progress

- **graph-TOKEN** fine-tune (latent fusion, 10 tokens incl. null_in/out) -- testing "latent > text" + whether it parrots less / transfers to test better.
- **CoT zero-shot ceiling** -- VLM reasons over scene / head-body orientation / interaction, then answers (semantic-reasoning capability without the graph).